Temporality promotes discontinuous transitions and bistability in higher-order contagion

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Higher-order contagion models show that group-based reinforcement can trigger discontinuous transitions, yet these results are typically derived on static hypergraphs that neglect temporal group dynamics. Here, we demonstrate that temporality is essential for correctly characterizing complex contagion by developing an analytically solvable approximate master equation framework that incorporates group turnover in the form of annealing. Our theory interpolates continuously between quenched and mean-field limits and reveals that increasing annealing can induce a discontinuous transition, enlarge the bistable region, and, under sufficiently strong nonlinearities or large group sizes, reverse its influence on the invasion threshold, hindering contagion. Mapping empirical annealing rates to theory further shows that real systems may exhibit either continuous or discontinuous transitions, with the latter requiring stronger non-linear reinforcement than previously thought, highlighting that simulations on static structures and mean-field analyses can both yield qualitatively misleading predictions.

In social systems, the adoption of new behaviors or ideas often requires more than a single exposure: individuals typically need reinforcement from multiple peers before changing state [1–3]. This feature has motivated a broad class of complex contagion models [4–12] traditionally based on threshold rules [4–9] or on nonlinear synergistic mechanisms [10–12]. Recent work by Iacopini et al. [13] revitalized this field by representing reinforcement through its natural substrate: groups rather than pairwise interactions. Their formulation recovered the pivotal result that collective reinforcement can shift the transition from a no-adopter to an adopter state from continuous to discontinuous [6, 12], and spurred developments across diverse dynamical processes, including synchronization [14–18], self-organization [19], and game theory [20–22]. Yet, despite these advances, current models provide little insight into how contagion unfolds when groups themselves evolve in time.

Real group interactions are inherently temporal [23]: individuals join and leave groups on time scales slower than, comparable to, or faster than contagion itself. However, mathematical descriptions of discontinuous transitions typically adopt the mean-field, fully annealed limit [13, 24], in which individuals reshuffle infinitely fast and effectively experience a perfectly mixed environment where all possible groups co-exist at all times. Under this assumption, Iacopini et al. [13] derive an invasion threshold that is independent of the nonlinearity of the contagion dynamics. However, when the temporal evolution of the structure is explicitly incorporated, this picture can change markedly. For linear contagion, temporality is known to lower the invasion threshold by enhancing mixing [25, 26], while in the opposite limit of a static (quenched) structure, the threshold can even diverge if each individual belongs to a single group only [27].

Surprisingly, the limited efforts aimed at understanding temporal complex contagion report a conflicting trend. Chowdhary et al. [28] showed through simulations that the threshold shifts towards larger values on time-varying simplicial complexes, regardless of whether contagion is linear or nonlinear. This is the opposite of what is well established for linear contagion [25, 26], raising the question of whether higher-order reinforcement behaves fundamentally differently or whether this discrepancy arises from how temporality is incorporated.

Complementing this theoretical gap, empirical analyses also highlight the importance of temporal structure: real group evolution is shaped by mechanisms such as memory, persistence, and group-size effects [29–31]. Yet, some studies validating discontinuous transitions using proximity datasets [13, 32, 33] typically simulate contagion on static, aggregated, representations

of these temporal streams [34–40]. This aggregation obscures group turnover and can distort structural features such as density and hyperedge overlap [18], which affect both the quantitative and qualitative outcomes of complex contagion dynamics [33, 41–43].

In this work, we quantify the temporal variability of group structure through the effective annealing rate, defined as the rate at which individuals reshuffle between groups across empirical datasets. We then develop an Approximate Master Equations (AMEs) framework that incorporates group annealing into complex contagion dynamics and solve it analytically. Using this model, we show: (*i*) how varying the annealing rate enables a smooth interpolation from the mean-field limit with infinite annealing to a quenched structure with zero annealing; (*ii*) how increasing annealing can transform a continuous phase transition into a discontinuous one and enlarge the bistability region for nonlinear contagion; (*iii*) how, for sufficiently strong nonlinearities and parallel exposures, or large group sizes, the effect of annealing on the invasion threshold reverses sign, with faster group turnover shifting the threshold towards larger values and hindering adoption; and (*iv*) how the temporal features of empirical data determine whether transitions appear continuous or discontinuous, demonstrating that static, aggregated representations may yield qualitatively misleading predictions.

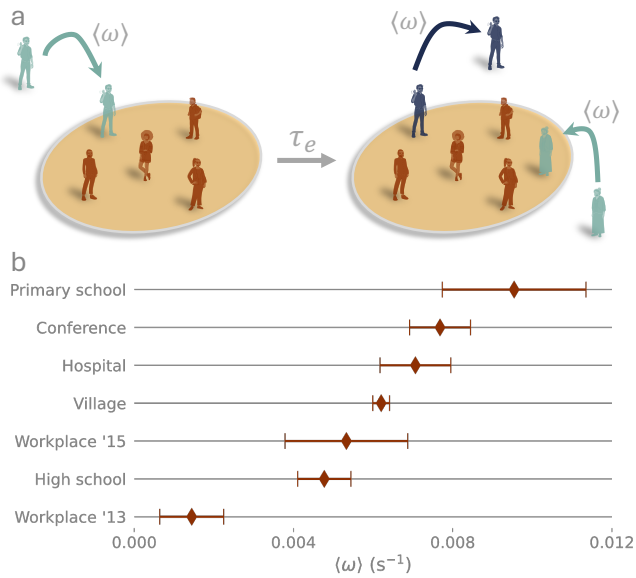

FIG. 1: **Quantifying the effective annealing rate in real interaction structures. a** Schematic illustration of membership turnover: individuals join and leave groups, generating a distribution of residence times τ_e. The effective annealing rate $\langle\omega\rangle$ summarizes the typical frequency of these group-change events. **b** Empirical values of $\langle\omega\rangle$ measured across several face-to-face interaction datasets, revealing variability in the rate of group reshuffling across social contexts.

Effective annealing rate in real interaction structures

Real human groups are not static entities: their composition evolves continuously through processes such as growth, contraction, merging, splitting, appearance and disappearance [23]. As recently shown by Iacopini et al. [29], most changes involve only a few individuals joining or leaving a group, and aggregation and disaggregation events occur at comparable rates. Across these diverse mechanisms, a common and dominant feature emerges: individuals are continually reallocated across groups, generating persistent membership turnover.

We capture this turnover through a single coarse-grained parameter, the effective annealing rate $\langle\omega\rangle$, which quantifies how frequently individuals change groups. Rather than modeling each microscopic reorganization event, $\langle\omega\rangle$ summarizes the typical rate of group reshuffling observed in the data. Conceptually, it represents the inverse of the average time an individual remains within the same group (see Fig. 1a and Methods for the precise definition).

Fig. 1b shows the empirical values of $\langle\omega\rangle$ across several face-to-face interaction datasets [34–40] (see Supplementary Note 1 and its Supplementary Fig. 2 for details on the structures, their preprocessing, and the group tracking). Reshuffling rates vary widely across contexts, spanning almost an order of magnitude, from more stable structures to rapidly evolving ones. This variability situates real systems at different points along the quenched–annealed spectrum and highlights that temporality is a fundamental characteristic of human group organization.

Minimal analytical framework for temporal higher-order contagion

Empirical interaction data reveal that individuals reshuffle between groups at characteristic rates (Fig. 1b). To capture the impact of this temporality on higher-order contagion, we construct a minimal analytical framework in which all changes in group composition are coarse-grained into a single node-level annealing process occurring at rate ω. We consider (*i*) a set of groups with size distribution $\{p_n\}$, where n denotes group size, and (*ii*) a population of individuals whose number of group memberships k follows the distribution $\{g_k\}$. Two processes govern the dynamics. *Complex contagion*: a susceptible individual in a group of size n with i adopters becomes an adopter at rate $\beta(n,i) = \lambda i^{\nu}$, where λ is the intrinsic adoption rate and ν con-

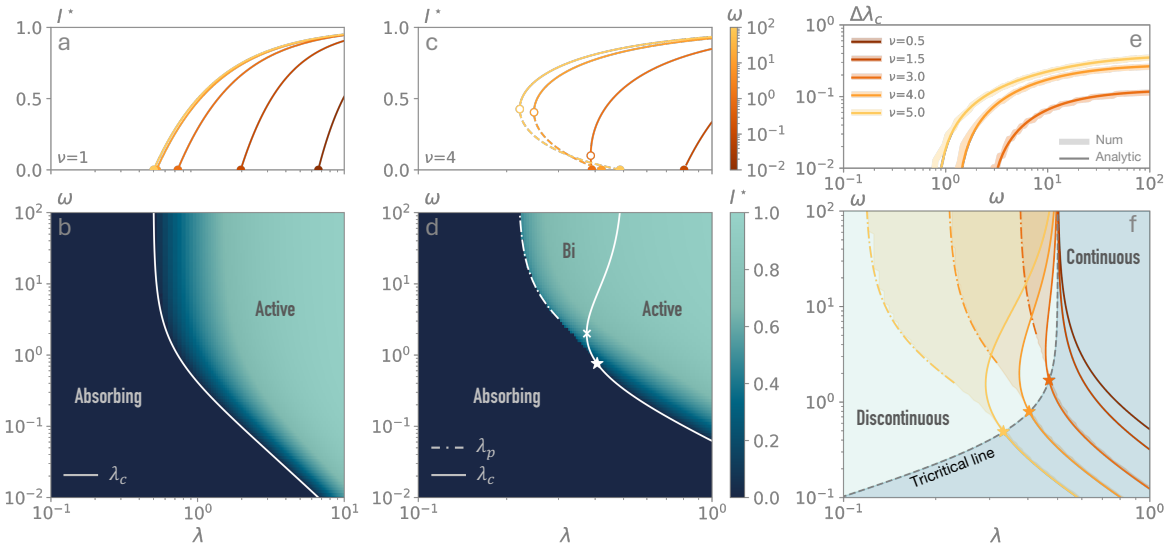

FIG. 2: **Temporality reshapes the critical behavior of higher-order contagion. a–b** Stationary prevalence $I^\star$ and corresponding phase diagram for linear contagion ($\nu = 1$) as a function of the spreading rate λ and the annealing rate ω. Increasing temporal turnover lowers the invasion threshold, and the transition remains continuous. **c–d** Same representation for nonlinear contagion ($\nu = 2.5$). Here, temporality fundamentally alters the transition: the invasion threshold becomes non-monotonic in ω, and sufficiently strong synergy produces a discontinuous transition. Analytical predictions for $\lambda_c(\omega)$ and $\lambda_p(\omega)$ are shown throughout, the tricritical point derived using Eq. (17) in Methods is marked with a star in panel d, and the minimum of the invasion threshold according to Eq. (4) is marked with a cross in panel d. **e** Width of the bistable region, $\Delta\lambda$, as a function of the annealing rate for different values of the synergy exponent ν. **f** Bifurcation diagram showing the analytically derived tricritical line separating continuous and discontinuous transitions, together with the corresponding invasion thresholds for the nonlinearities considered in panel e. All analytical curves closely match the steady-state solutions of the AMEs. In all panels, $k = 1$ and $n = 3$.

trols the strength of synergistic reinforcement; adopters recover at rate μ. *Annealing*: individuals reshuffle between groups at rate ω.

The resulting coevolutionary dynamics are described by a set of Approximate Master Equations (AMEs; Eqs. (9)-(10) in Methods). These equations track the density of susceptible individuals with membership k, s_k, and the fraction of groups of size n containing i adopters, $f_{n,i}$. From these quantities, we compute the stationary adopter population $I^\star$ (see Eq. (13) in Methods), which we use throughout the following sections to characterize the macroscopic behavior of temporal higher-order contagion.

Temporality induces non-monotonic thresholds and promotes bistability

Temporality fundamentally reorganizes the phase transition of higher-order contagion. In the sociologically grounded setting where individuals can effectively participate in only one face-to-face interaction at a time ($k = 1$ in our AMEs framework), temporal turnover reshapes the macroscopic behavior. As we show below, increasing the annealing rate does not merely shift the transition: it first creates a continuous transition that does not exist in the quenched limit, and then, for sufficiently strong nonlinear reinforcement, transforms it into a discontinuous transition with bistability.

In the quenched limit, $\omega = 0$, group composition is frozen, and individuals remain confined to their initial interaction patterns. Under these conditions, contagion cannot invade: adopters are not able to sustain activity, and the system always collapses into the absorbing state $I^\star = 0$ [27]. The invasion threshold therefore diverges (see Supplementary Notes 2,3 and 4 for the derivation),

$$\lambda_c(w \to 0) = \infty, \tag{1}$$

revealing that no transition exists in static higher-order networks with $k = 1$.

Any degree of temporal evolution ($\omega > 0$) changes this picture qualitatively. Even slow reshuffling exposes individuals to multiple group configurations over time. As a result, a transition from the absorbing to the active phase emerges. In the opposite limit of infinitely fast reshuffling ($\omega \to \infty$), individuals effectively sample all possible group realizations, yielding the mean-field critical point

$$\lambda_c(w \to \infty) = \frac{\mu \langle n \rangle}{\langle k \rangle \langle n(n-1) \rangle}, \tag{2}$$

which holds for both linear and nonlinear contagion (see Supplementary Note 4 for the derivation). The full curve interpolating between these limits is derived in the Supplementary Note 3 and detailed in Eq. (16) in Methods, and matches the steady-state AME solutions in Fig. 2a–d (see Methods for details in the computation).

Having established that temporality creates a transition that is absent in the quenched limit, we next examine how it changes the nature of this transition. When contagion is linear ($\nu = 1$), the annealing-induced transition is always continuous: $I^\star$ grows smoothly from zero as λ crosses λ_c. However, in the presence of nonlinear reinforcement, temporality can fundamentally alter the macroscopic behavior. As shown in Fig. 2c-d, for sufficiently strong synergy, increasing the annealing rate causes the continuous transition to give way to a discontinuous one accompanied by bistability, in which both the absorbing and active states are locally stable. The tricritical condition producing the star point in Fig. 2d is derived in the Supplementary Note 3 and is provided explicitly in Eq. (17) in Methods. As shown in Fig. 2e-f, the width of the bistable region increases with both the annealing rate ω and the synergy exponent ν, revealing that temporal turnover amplifies the cooperative effects needed to generate explosive adoption events.

Once this change in the transition's nature is understood, a second and equally striking consequence of temporality becomes clear: in the nonlinear regime, the invasion threshold $\lambda_c(\omega)$ becomes non-monotonic in the annealing rate. Fig. 2c–d show that $\lambda_c(\omega)$ initially decreases as temporal mixing increases, reaching a finite minimum at $\omega^\star$, after which it increases again. Thus, temporality enhances contagion only up to an optimal mixing rate; beyond this point, additional reshuffling suppresses the reinforcement required for adoption.

This non-monotonicity is a hallmark of higher-order interactions. When only pairwise interactions are present ($n = 2$), the critical line reduces to $\lambda_c(\omega) = \mu + \mu^2/\omega$, which is strictly monotonic and independent of ν. In contrast, for structure consisting only of triplets ($n = 3$), the critical line as derived in the Supplementary Note 4 reads

$$\lambda_c(\omega) = \frac{\mu + \omega}{2^\nu}\left[\sqrt{1 + \frac{2^\nu \mu}{\omega}} - 1\right], \tag{3}$$

from which a finite minimum emerges only when $\nu > 2$. The corresponding optimal mixing rate and minimal threshold are

$$\omega^\star = \mu\,\frac{2^{\nu/2}}{2^{\nu/2} - 2}, \qquad \lambda_c^\star = \mu\,\frac{2^{\nu/2} - 1}{2^{\nu-1}}. \tag{4}$$

In contrast, the prevalence threshold, i.e., the left boundary of the bistability region, remains strictly monotonic in ω. For large annealing rates and $n = 3$, we derive in the Supplementary Note 5 the closed expression reported in Eq. (18) in Methods; its asymptotic value is

$$\lambda_p(\omega \to \infty) = \mu\frac{4\left(2^\nu - 2\right)}{2^{2\nu}}, \tag{5}$$

and therefore, accounting for Eq. (2), the width of the bistable region approaches

$$\Delta\lambda(\omega \to \infty) = \frac{\mu}{2}\left(1 - 2^{2-\nu}\right)^2. \tag{6}$$

As shown in Fig. 2, these analytical expressions match the stationary solutions of the AMEs.

Together, these results show that temporality has a dual role. It first enables contagion to spread by creating a continuous transition, and subsequently it intensifies cooperative effects to the point of generating a discontinuous transition with bistability. At the same time, the critical threshold itself becomes non-monotonic, displaying an optimal turnover rate at which the system is most vulnerable to adoption.

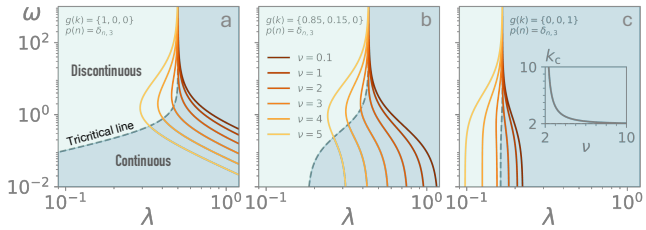

FIG. 3: **Interplay between connectivity, annealing, and synergy reshapes the phase diagram.** Phase portraits showing the critical adoption threshold λ_c as a function of the annealing rate ω_c for different membership distributions: $\langle m \rangle = 1.00$, 1.15, and 3.00 respectively for panels **a**, **b** and **c**. Each panel highlights the tricritical lines separating the regions with continuous and discontinuous transitions. The inset in panel **c** shows the critical membership needed to erase the non-monotonicity in the invasion threshold according to Eq. (7)

Parallel exposure reverses the effect of temporality in nonlinear contagion

Up to this point we have focused on the case $k = 1$, where individuals participate in only one group at a time. However, modern social settings, especially those involving digital communication, allow individuals to receive multiple stimuli in parallel. In our framework, this corresponds to increasing the effective membership k, and Fig. 3 illustrates how this reshapes the phase portrait for fixed group size $n = 3$.

For nonlinear contagion, increasing k progressively alters the influence of temporality. When $k = 1$ (see 2

and 3a), the invasion threshold $\lambda_c(\omega)$ exhibits the non-monotonic behavior described in the previous section: it decreases with ω, reaches a finite minimum, and then increases again. When some individuals can participate in two groups (Fig. 3b), this minimum shifts toward $\omega = 0$. For $k = 3$ (Fig. 3c), the minimum disappears for sufficiently large nonlinearities, and the threshold becomes strictly increasing in ω: temporal reshuffling now consistently hinders adoption by disrupting the stability needed for reinforcement to accumulate.

This qualitative change is governed analytically by the critical membership

$$k_c(\nu) = \frac{2^{\nu/2+1} + 2^{\nu+1} - 4}{2^{\nu} - 4}, \tag{7}$$

derived from the explicit expression for $\lambda_c(\omega, k)$ (see Eq. (22) in Methods and Supplementary Note 4 for the derivation) and shown in the inset of Fig. 3c. For $k < k_c(\nu)$, the threshold remains non-monotonic and displays an optimal turnover rate (see Eqs. (23)-(24) in Methods and Supplementary Note 4 for the derivation). For $k > k_c(\nu)$, $\lambda_c(\omega, k)$ becomes strictly increasing: temporality always suppresses contagion. Since $k_c(\nu)$ approaches a horizontal asymptote in $k = 2$, even modest parallel exposure is enough to reverse the role of annealing when synergy is strong.

This reversal is not limited to the case $n = 3$. As derived in Supplementary Note 6 and shown in Supplementary Fig. 2, increasing the typical group size n amplifies higher-order reinforcement and reshapes the monotonicity of the invasion threshold. Larger groups make non-monotonicity emerge at smaller synergy values and push the minimum of $\lambda_c(\omega)$ to negative annealing rates, thereby erasing it within the physical domain $\omega > 0$. Consequently, when n is sufficiently large, $\lambda_c(\omega)$ becomes strictly decreasing even for moderate and low synergy ν and small membership k. Increasing n also weakens the influence of annealing on the order of the transition, although less strongly than parallel exposure. Together, these results demonstrate that parallel exposure fundamentally reshapes how temporality affects higher-order contagion: while temporal mixing facilitates linear spreading, it can hinder nonlinear contagion once reinforcement becomes strong, reversing classical expectations.

Temporal reshuffling determines the transition class in empirical systems

Having established how temporality reshapes higher-order contagion in synthetic settings, we now examine real social systems. For each dataset, we infer three empirical ingredients: the group-size distribution $p(n)$, the membership distribution $g(k)$, and the effective annealing rate $\langle\omega\rangle$ (see Supplementary Fig. 1 in Supplementary Note 1, and Fig. 1). Within the AMEs framework, these quantities fully determine the tricritical point $(\lambda_c^\star/\mu, \nu_c^\star)$ that separates continuous from discontinuous transitions.

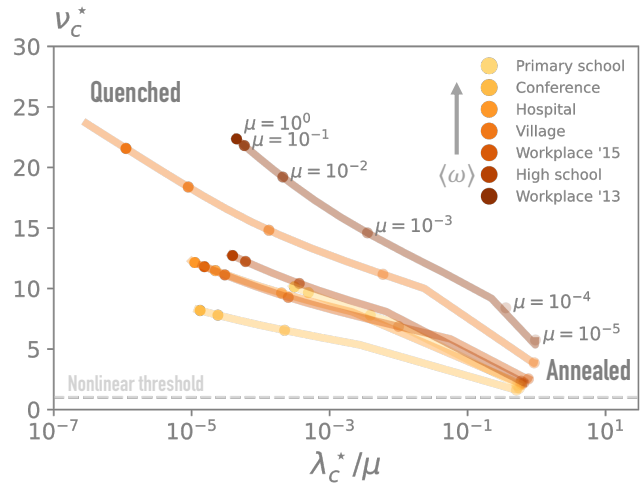

FIG. 4: **Critical boundary as a function of the annealing rate across dynamical timescales.** Empirical critical points $(\lambda_c^\star/\mu, \nu_c^\star)$ for six intrinsic timescales of the contagion dynamics, from $\mu = 10^0$ to $\mu = 10^{-5}$. For each dataset, coloured curves show how the critical point moves as the annealing rate ω is swept from the quenched to the strongly annealed regime; markers indicate the position corresponding to the empirically measured annealing rate $\langle\omega\rangle$. Large μ (fast contagion) makes temporal evolution negligible, preserving co-membership and maintaining high nonlinear thresholds, whereas small μ (slow contagion) suppresses reinforcement and collapses the system toward the annealed regime. Note that the color of the datasets indicates the order of the effective annealing rate from lowest to highest.

Fig. 4 summarises the resulting empirical phase boundaries. Each coloured curve shows how the critical point moves as the annealing rate ω is swept from the quenched to the annealed regime. The markers on each curve correspond to the empirically observed value $\langle\omega\rangle$ for a given contagion timescale μ^{-1}, indicating where each real system lies along this continuum.

A clear pattern emerges. When contagion is fast ($\mu \to \infty$), individuals recover quickly compared to structural reshuffling, and the interaction pattern appears effectively static. Reinforcement persists and tricritical points occur at large nonlinearities $\nu_c^\star$, characteristic of a quenched regime. When contagion is slow ($\mu \to 0$), the population reshuffles many times during the adoption period, erasing reinforcement and driving systems toward the annealed limit, where $\nu_c^\star$ becomes small.

Across datasets, the effective annealing rate $\langle\omega\rangle$ is therefore esential for determining whether the transi-

tion is continuous, discontinuous, or absent. Systems with slow turnover preserve reinforcement and remain closer to the quenched regime, whereas fast-turnover environments operate near the annealed limit. Notably, achieving a discontinuous transition in any empirical setting requires consistently large nonlinearities, highlighting that strong reinforcement is necessary for bistability to emerge.

Together, these results show that the relative timescales of contagion and social reshuffling govern the nature of the critical boundary. Temporality is thus a primary determinant of the transition class of higher-order contagion, and static structures entirely miss this dimension.

Discussion

Temporality profoundly reshapes collective dynamics on higher-order structures. We introduced a simplified yet empirically grounded description of group turnover in the form of a single annealing rate, which captures the rate of changes in group membership while preserving the heterogeneity of the group sizes and membership distributions. Using this solvable model, we uncovered how temporal variability in group composition reorganizes the entire phase diagram of complex contagion. Our analytical framework shows that the relative timescales of contagion and membership reshuffling determines whether the macroscopic transition is continuous, discontinuous, or altogether absent. This is an effect invisible to both static hypergraphs and fully annealed approximations.

A central outcome of this work is the reconciliation of seemingly contradictory results. Classical linear contagion predicts that temporal mixing lowers invasion thresholds [25, 26], whereas simulations on temporal simplicial complexes reported the opposite trend even in the linear contagion case [28]. Our theory clarifies that both behaviors arise, but in different dynamical regimes: annealing facilitates spreading for linear or weakly nonlinear contagion, but for sufficiently strong synergy the mechanism reverses. Fast reshuffling prevents individuals from remaining long enough within stable groups to accumulate reinforcement, thereby hindering adoption. This reversal emerges robustly across group sizes, membership distributions, and empirical datasets. The opposite trend observed by Chowdhary et al. [28] is explained by the fact that their temporal reshuffling does not preserve the degree sequence, or other statistical heterogeneity, effectively comparing quenched heterogeneous networks to annealed homogeneous networks (see Supplementary Fig. 3 in Supplementary Note 7).

Our results also reveal that discontinuous transitions require far stronger nonlinear reinforcement than previously appreciated. Across all datasets examined, the nonlinearities needed to observe bistability are consistently high. This underscores that claims derived from dynamics on static, aggregated hypergraphs, where reinforcement is artificially inflated, should be interpreted with caution.

More broadly, our findings show that temporality is not a correction to network structure but a primary dynamical ingredient. Important effects of temporality have also been observed in synchronization and consensus dynamics, where time-varying interactions accelerate convergence [44, 45]. Here, we demonstrate that temporality also determines the qualitative nature of higher-order contagion: systems fall into distinct dynamical regimes depending on the balance between group reshuffling, group size, nonlinearity, and the effective membership.

Several avenues for future work emerge. First, the annealing process itself may be adaptive: individuals modify their group participation in response to contagion [19] or in response to group traits [46, 47], feedback that could be incorporated into solvable extensions of the present AMEs. Second, combining higher-order contagion with temporal heterogeneity in transmission, memory, or behavioural responses may yield richer classes of dynamical transitions. Finally, our results raise methodological implications: aggregated hypergraphs can significantly misrepresent dynamical behaviour, and future empirical studies should integrate temporal information whenever possible.

Together, these results establish a unified framework for temporal higher-order contagion, and reveal that temporality is not a secondary correction to higher-order contagion, but a primary mechanism that determines the macroscopic transition class. Ignoring temporal turnover fundamentally mischaracterizes the nature and location of the transition (whether it unfolds smoothly, explosively, or not at all), highlighting the need to treat time as an intrinsic part of the interaction substrate.

METHODS

Definition of the effective annealing rate

To quantify this reshuffling process, we measure the average residence time of individuals within groups. Let τ_e denote the time interval between two consecutive group-change events involving the same individual (see Fig. 1a), and let E be the total number of such events in the dataset. We define the effective annealing rate as

$$\langle \omega \rangle = \frac{1}{\langle \tau \rangle}, \qquad \langle \tau \rangle = \frac{1}{E} \sum_{e=1}^{E} \tau_e \tag{8}$$

which corresponds to viewing group departures as a Poisson process. This quantity provides a direct measurement of how rapidly group composition changes in real systems. For details in the computation of the error in Fig. 1, we refer the reader to the Supplementary Note 1.

wAMEs model of complex contagion with annealing

We follow the framework of St-Onge et al. [32], a non-linear generalization of the group-based Approximate Master Equations (AMEs) [27]. We add a mechanism for temporal annealing at rate ω and track the evolution of the susceptible population with membership k, denoted s_k, and the fraction of groups of size n with i infectious individuals within, denoted $f_{n,i}$. The governing equations are

$$\frac{ds_k}{dt} = \mu(1-s_k) - krs_k \tag{9}$$

$$\begin{aligned}\frac{df_{n,i}}{dt} &= (i+1)\big(\mu+\omega(1-I)\big)\, f_{n,i+1} \\ &\quad - \left[i(\mu+\omega(1-I)) + (n-i)(\beta(n,i)+\rho+\omega I)\right] f_{n,i} \\ &\quad + (n-i+1)(\beta(n,i-1)+\rho+\omega I) f_{n,i-1}.\end{aligned} \tag{10}$$

Equation (9) describes the change in the fraction of susceptible individuals with k group memberships. This fraction increases when infected individuals recover and decreases when susceptibles become infected through exposure in any of the k groups to which they belong.

The terms in Eq. (10) capture all transitions that modify the number of infectious individuals within a group of size n. The first term represents transitions from state $i+1$ to i, resulting either from the recovery of an infectious individual or from its being swapped with a susceptible one. The second term accounts for all events that remove probability mass from $f_{n,i}$: any of the i infectious individuals may recover or be swapped out, and any of the $n-i$ susceptible individuals may become infected through within-group transmission, external influence ρ, or by being swapped with an infectious individual. The third term describes transitions from state $i-1$ to i, occurring when one of the $n-(i-1)$ susceptibles becomes infectious via internal transmission, external influence, or by being swapped with an infectious individual. Importantly, the external influence ρ represents the mean-field infection pressure exerted by all groups to which a susceptible node belongs, excluding the focal group, and is given by

$$\rho(r) = r \cdot \frac{\sum_k k(k-1)s_k g_k}{\sum_k k s_k g_k}, \tag{11}$$

which corresponds to the product of the mean-field infection rate coming from a randomly selected external group and the mean excess membership of a susceptible node. In other words, if we pick a susceptible node in a given group, the prefactor represents the expected number of other groups to which it belongs. The mean-field infection rate itself is defined as

$$r = \frac{\sum_{n,i} \beta(n,i)(n-i) f_{n,i} p_n}{\sum_{n,i}(n-i) f_{n,i} p_n}, \tag{12}$$

which is the average value of $\beta(n,i)$ times the susceptible members within a group over the distribution of infected nodes. In addition, the global prevalence,

$$I(t) = \sum_k (1 - s_k(t) g_k), \tag{13}$$

serves as the main order parameter of the system. Together, these contributions provide a compact characterization of the gain and loss processes that govern the evolution of $f_{n,i}$.

Stationary dynamics and computation of the stationary states

In the stationary state, the whole system can be rewritten entirely in terms of the variables r, ρ and I (See Supplementary Note 2). We therefore define the functional

$$\mathcal{M}\left[\rho(r), I(r)\right] = \frac{\sum_{n,i} \beta(n,i)(n-i) f_{n,i}(\rho, I) p_n}{\sum_{n,i}(n-i) f_{n,i}(\rho, I) p_n}, \tag{14}$$

where the dependency on ρ and I enters through the stationary group-state distributions $f_{n,i}(\rho, I)$. At equilibrium, the system must satisfy the implicit self-consistency condition

$$r = \mathcal{M}\left[\rho(r), I(r)\right]. \tag{15}$$

This relation can be solved numerically to obtain the fixed points of the dynamics. To determine their stability, we additionally evaluate the Jacobian of the stationary system, as detailed in the Supplementary Note 2.

Invasion threshold

The critical value for the invasion threshold can be derived by imposing that $\frac{\partial M}{\partial r}\big|_{r\to 0} = 1$. As derived in the Supplementary Note 3, the condition reads

$$\begin{aligned}1 \;=\; &\left\langle \sum_{i=1}^{n} \frac{n!}{(n-i-1)!\, i!} \left(\frac{\lambda_c}{\mu+\omega}\right)^i \prod_{j=1}^{i} j^{\nu} \right\rangle \\ &\times \left(\frac{\langle k(k-1)\rangle}{\langle k\rangle\langle n\rangle} + \frac{\omega}{\mu}\frac{\langle k\rangle}{\langle n\rangle} \right).\end{aligned} \tag{16}$$

Bistability threshold

To fully characterize the phase portrait, we derive in the Supplementary Note 3 the condition for the tricritical point that separates the absorbing, active, and bistable regions. Imposing that $\left.\frac{\partial^2 M}{\partial r^2}\right|_{r\to 0} = 0$ we reach that the condition reads

$$0 = F\frac{\langle k(k-1)\rangle^2}{\langle k\rangle^2} + 2G\frac{\langle k(k-1)\rangle}{\langle k\rangle}\frac{\langle k\rangle}{\mu} + H\frac{\langle k\rangle^2}{\mu^2} + 2\frac{\langle k^2\rangle^2 - \langle k^3\rangle\langle k\rangle - \frac{w}{\mu^2}\langle k^2\rangle\langle k\rangle^2}{\langle k(k-1)\rangle\langle k\rangle + \frac{w}{\mu}\langle k\rangle^3}. \quad (17)$$

Here, $F \equiv F\left[p_n, \beta(n,i), w\right]$, $G \equiv G\left[p_n, \beta(n,i), w\right]$, $H \equiv H\left[p_n, \beta(n,i), w\right]$ are specified in the Supplmentary Note 2.

Persistance threshold

The expression of the persistance threshold $\lambda_p(\omega)$ for large values of ω is obtained in the Supplementary Note 5, and reads

$$\lambda_p(\omega) = \frac{-B_p(\omega) + \sqrt{B_p(\omega)^2 - 4A_p(\omega)C_p(\omega)}}{2A_p(\omega)}, \quad (18)$$

where

$$A_{\rm p}(\omega,\mu) = 2^{\nu+1}\big(2^\nu\omega + 3\mu\big), \quad (19)$$

$$B_{\rm p}(\omega,\mu) = 4^\nu\omega^2 + 2^{\nu+1}\mu\,\omega + 8\mu\,\omega + 9\mu^2, \quad (20)$$

$$C_{\rm p}(\omega,\mu) = -4\big(2^\nu - 2\big)(\omega+\mu)^2. \quad (21)$$

Optimal annealing rate to maximize adoption

The general expression for the critical line is derived in the Supplementary Note 4 and reads

$$\lambda_c(\omega;k) = \frac{\mu+\omega}{2^\nu}\left[\sqrt{1 + \frac{2^\nu\mu}{k\omega + \mu(k-1)}} - 1\right], \quad (22)$$

which reduces to Eq. (16) for $k = 1$. For $\nu > 2$, differentiating Eq. (22) yields the position of the minimum:

$$\omega^\star(k) = \mu\,\frac{2^{\nu/2+1} + 2^{\nu+1} - 4 - k(2^\nu - 4)}{k(2^\nu - 4)}, \qquad (\nu > 2), \quad (23)$$

which reduces to $\omega^\star = \mu\, 2^{\nu/2}/(2^{\nu/2} - 2)$ for $k = 1$. As k increases, the minimum $\omega^\star(k)$ shifts toward zero and eventually becomes non-positive. When this happens, the minimum disappears entirely and $\lambda_c(\omega;k)$ becomes strictly increasing in ω.

Evaluating Eq. (22) at $\omega^\star(k)$ gives the minimal threshold

$$\lambda_c^\star(k) = \frac{1}{k}\,\lambda_c^\star(1) = \frac{\mu}{k}\,\frac{2^{\nu/2} - 1}{2^{\nu-1}}. \quad (24)$$

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